

Amit Kumar Mohanty

mohantyamit2003@gmail.com | +91 9354937256 | github.com/amitmohanty022 | linkedin.com/in/amitkrmohanty

Target Role: AI Engineer | Keywords Studios

PROFESSIONAL SUMMARY

AI Engineer with 1+ year of hands-on experience building, deploying, and optimizing Generative AI and Large Language Model (LLM) solutions in production. Specialized in context-aware, autonomous AI agents and multi-agent orchestration, RAG pipelines, and conversational AI served via scalable FastAPI REST APIs. Strong Python engineer across the modern LLM ecosystem (transformers, PEFT/LoRA), skilled in fine-tuning foundation models, prompt engineering (Few-Shot, Chain-of-Thought, ReAct), multimodal AI, and LLMops (Docker, CI/CD) on AWS and GCP. Peer-reviewed research author focused on turning state-of-the-art AI into reliable products that impact millions of users.

EDUCATION

K.R. Mangalam University, Gurgaon

July 2021 - July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML)

Birla Institute of Technology and Science, Pilani (BITS Pilani)

July 2026 - July 2028

M.Tech in Artificial Intelligence and Machine Learning (Pursuing)

EXPERIENCE

Research Associate - AI, Keywords Studios India - Gurgaon, Haryana

May 2025 - Present

- Built context-aware, autonomous AI agents and multi-agent workflows using modern frameworks to automate complex business logic and multi-step tasks.
- Designed, trained, and deployed production-ready GenAI pipelines; collaborated with data engineers to preprocess and curate large datasets, delivering 85% of training data to production.
- Experimented with state-of-the-art LLMs and fine-tuned foundation models, running structured evaluations to optimize model performance; awarded "Rookie of the Year 2025."

Artificial Intelligence Intern, Infosys - Remote

Nov 2024 - Feb 2025

- Built conversational AI with LangChain and LLM orchestration, raising response accuracy by 35% through advanced prompt engineering (Few-Shot, Chain-of-Thought).
- Defined evaluation and guardrails that reduced hallucinations by 20%, and optimized model deployment for latency (-25%) within a 12-engineer Agile team.

PROJECTS

Dynamic Screen Companion - Multimodal LLM + RAG

June 2025 - Present

- Built a real-time multimodal (vision + language) GenAI assistant integrating LLM APIs (Gemini/OpenAI) with a RAG retrieval layer for context-aware responses.
- Engineered a FastAPI REST backend serving concurrent inference streams; built the embeddings + vector-search pipeline and containerized with Docker + CI/CD.

Diabetic Optiscan - Vision Transformer (Peer-reviewed research)

Jan 2025 - June 2025

- Built and fine-tuned a Vision Transformer (ViT-B/16) on the APTOS dataset; co-authored a peer-reviewed paper on the architecture and accuracy gains.
- Created a custom wavelet technique improving detection precision by 50% over legacy methods.

SKILLS

Languages & LLM Ecosystem: Python, Hugging Face Transformers, PEFT/LoRA, QLoRA, accelerate, PyTorch, TensorFlow, scikit-learn, NumPy, Pandas

LLMs & Orchestration: LLMs (Gemini, GPT-4, Claude), LangChain, LangGraph, LlamaIndex, OpenAI SDK, Google AI SDK, Prompt Engineering (Few-Shot, Chain-of-Thought, ReAct)

Agentic AI: Autonomous AI Agents, Context-Aware Agents, Multi-Agent Systems, Agent Orchestration, Tool Use, Workflow Automation

RAG, Vectors & Data: RAG Pipelines, Embeddings, Vector Databases (FAISS, Chroma, Pinecone), Data Preprocessing & Curation, SQL, MongoDB

APIs & MLOps: FastAPI, Flask, RESTful APIs, Model Serving, Docker, Kubernetes, CI/CD, MLflow, Deployment & Monitoring

Deep Learning & Cloud: Fine-tuning Foundation Models, Multimodal AI (Vision + Language), Transformers, NLP, Computer Vision, AWS, GCP (Vertex AI), Azure

CERTIFICATIONS

- Peer-reviewed research publication in medical AI (Vision Transformers for diabetic retinopathy detection).
- Ducat India - Data Science Professional Training (Python, ML, statistics, predictive modeling), Apr 2024 - Apr 2025.
- ISRO - AI/ML for Geodata Analysis (applied ML/DL to remote-sensing imagery).